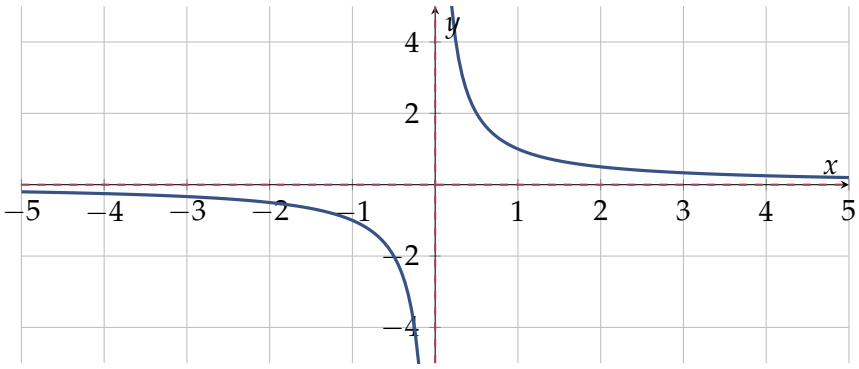
CHAPTER 4

Reciprocal (inverse) function

Skills: finding a domain, differentiating, studying a sign, drawing up a table of variations, graphing

4.1 Reference notes

The reciprocal function is defined by $f(x) = \dfrac{1}{x}$ on $\mathbb{R} \setminus \{0\}$.



Key point. $f'(x) = -\dfrac{1}{x^2} < 0$ for every $x \neq 0$. The reciprocal function is strictly decreasing on each of the intervals $]-\infty, 0[$ and $]0, +\infty[$. The lines $x = 0$ and $y = 0$ are asymptotes.

Watch out. One cannot write that the reciprocal function is decreasing "on $\mathbb{R} \setminus \{0\}$" without specifying the two intervals. For example $-1 < 1$ but $f(-1) = -1 < 1 = f(1)$, which would contradict the global definition of decreasing.

For a function $g(x) = P(x) + \dfrac{k}{x}$, proceed in this order: domain; derivative; common denominator and factorisation; sign of the derivative; variations; limits or behaviour at the boundaries; useful points; graph.

Example 1. On $]0, +\infty[$, $g(x) = x + \dfrac{4}{x}$ satisfies $g'(x) = 1 - \dfrac{4}{x^2} = \dfrac{(x-2)(x+2)}{x^2}$. The denominator is positive; g' is negative on $]0, 2[$ then positive on $]2, +\infty[$. The function has a minimum $g(2) = 4$.

Session 23 — Definition and graph of the reciprocal function

Reciprocal (inverse) function

15 Dec 2026 • B: 2 h – C: 1 h

Objectives.

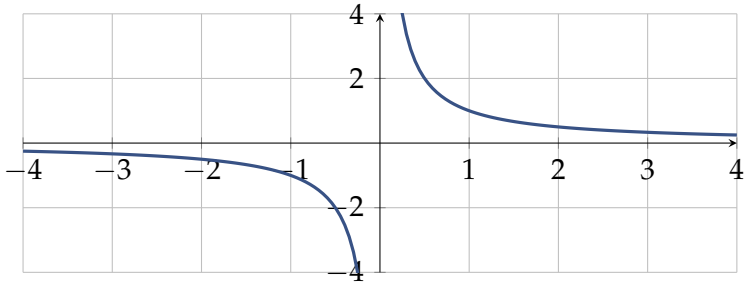
- Find the domain.
- Draw the curve.
- Use symmetries.

Fluency · 6 to 8 minutes, no calculator.

1. Simplify $3x + 5 - x + 2$. $2x + 7$
2. Give the excluded value of $\dfrac{x+1}{x}$. $x = 0$
3. Calculate $\dfrac{1}{5}$ and $\dfrac{1}{-5}$. $\dfrac{1}{5}$ **and** $-\dfrac{1}{5}$
4. Differentiate $3x^2 - 5x + 1$. $6x - 5$
5. Solve $(x-2)(x+1) = 0$. $x = -1$ **or** $x = 2$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: find the domain and draw the curve.

Activity 1. Calculate $\dfrac{1}{x}$ for $x = -4, -2, -1, -\dfrac{1}{2}, \dfrac{1}{2}, 1, 2, 4$. Plot the points and describe the symmetries. Why is $x = 0$ missing?



Solution. The values are respectively $-\frac{1}{4}, -\frac{1}{2}, -1, -2, 2, 1, \frac{1}{2}, \frac{1}{4}$. The domain excludes 0; the curve is symmetric about the origin. Have students say aloud which piece of data triggers the choice of method.

Written record. The reciprocal function is $f(x) = \dfrac{1}{x}$ on $\mathbb{R} \setminus \{0\}$. Its curve is a hyperbola symmetric about the origin.

Example 2. Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Solution. The values are respectively $-\frac{1}{4}, -\frac{1}{2}, -1, -2, 2, 1, \frac{1}{2}, \frac{1}{4}$. The domain excludes 0; the curve is symmetric about the origin.

Exercise 4.1 · essential.

1. Give the domain and the asymptotes of $f(x) = 5 + \dfrac{7}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study the function $g(x) = x + \dfrac{25}{x}$ on $]0, +\infty[$.

Solution. The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 5$. $f'(x) = -\dfrac{7}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{25}{x^2}$ and the minimum is reached at $x = 5$, with value 10.

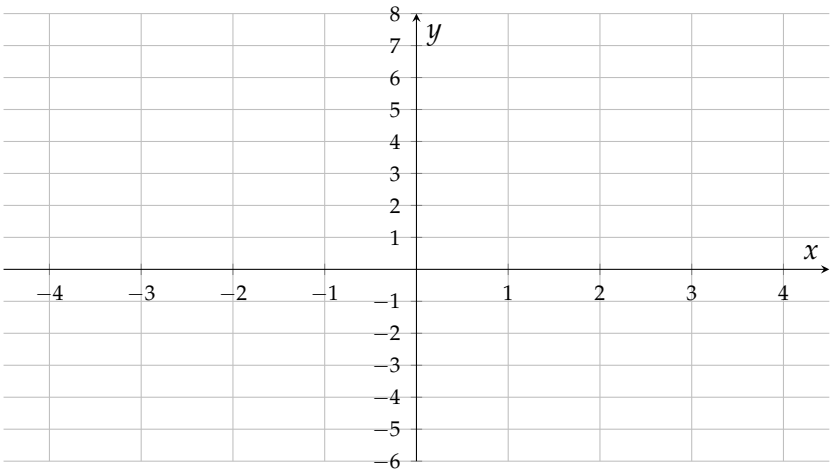
Further practice: Calao, course pp. 88–93; exercises p. 94, no. 1–5. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. Consider the functions f and g defined on $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$ by $f(x) = -\dfrac{2}{x}$ and $g(x) = 1 + \dfrac{3}{x}$.

1. Explain why 0 does not belong to the domain of these functions.
2. Complete the table of values (values rounded to the nearest hundredth if necessary).

x	-4	-3	-2	-1	-0.5	0.5	1	2	3	4
$f(x)$										
$g(x)$										

3. Plot the points and draw the curves $\mathcal{C}_f$ and $\mathcal{C}_g$ on the grid below.



4. Show that, for every non-zero real x, $f(-x) = -f(x)$. Which symmetry of $\mathcal{C}_f$ does this express? For each curve, give the lines it gets closer and closer to (asymptotes).

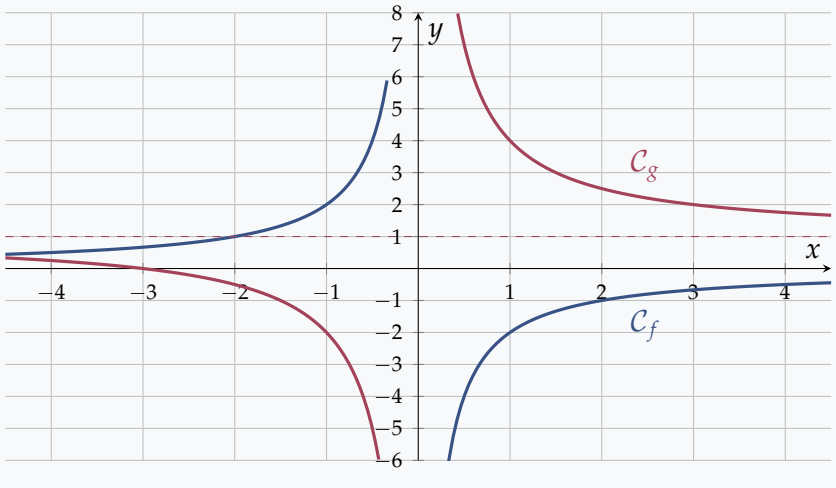
Solution. **Target:** Table of values and plotting hyperbolas.

1. Computing $\frac{2}{x}$ or $\frac{3}{x}$ requires dividing by x; division by 0 is impossible. The domain is $\mathbb{R}^* = (-\infty, 0) \cup (0, +\infty)$.

2.

x	-4	-3	-2	-1	-0.5	0.5	1	2	3	4
$f(x)$	0.5	0.67	1	2	4	-4	-2	-1	-0.67	-0.5
$g(x)$	0.25	0	-0.5	-2	-5	7	4	2.5	2	1.75

3.



4. $f(-x) = -\frac{2}{-x} = \frac{2}{x} = -f(x)$. The points with coordinates $(x, f(x))$ and $(-x, -f(x))$ are symmetric about the origin: $\mathcal{C}_f$ is symmetric about the origin O. Asymptotes: for $\mathcal{C}_f$, the lines with equations $x = 0$ and $y = 0$; for $\mathcal{C}_g$, the lines with equations $x = 0$ and $y = 1$ (the curve of g is symmetric about the point $(0, 1)$, since $g(-x) + g(x) = 2$).

Homework. Redo practice exercise 2 with the following data: replace the initial value by 235 and the rate or common ratio by 7. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The answer depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 24 — Behaviour at the boundaries and asymptotes

Reciprocal (inverse) function

17 Dec 2026 • B: 1 h – C: 2 h

Objectives.

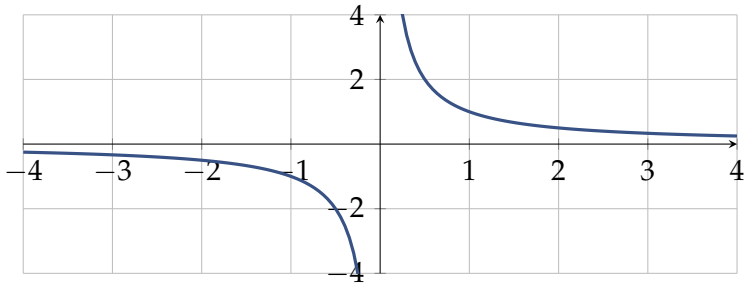
- Describe the behaviour at the boundaries.
- Identify the asymptotes.
- Read a graph.

Fluency · 6 to 8 minutes, no calculator.

1. Calculate 5% of 120. 6
2. Calculate $\frac{1}{6}$ and $\frac{1}{-6}$. $\frac{1}{6}$ **and** $-\frac{1}{6}$
3. Differentiate $3x^2 - 5x + 1$. $6x - 5$
4. Solve $(x-2)(x+1) = 0$. $x = -1$ **or** $x = 2$
5. Give the excluded value of $\frac{x+1}{x}$. $x = 0$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: describe the behaviour at the boundaries and identify the asymptotes.

Activity 2. Using a table of values near 0 and for large values of $|x|$, describe $\frac{1}{x}$. Translate each behaviour into an asymptote and a sentence.



Solution. Near 0, the values become very large in absolute value: asymptote $x = 0$. As $|x|$ grows, $\frac{1}{x}$ gets closer to 0: asymptote $y = 0$. Have students say aloud which piece of data triggers the choice of method.

Written record. For $f(x) = \frac{1}{x}$, $x = 0$ is a vertical asymptote and $y = 0$ a horizontal asymptote. The branches approach them without crossing them.

Example 3. Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Solution. Near 0, the values become very large in absolute value: asymptote $x = 0$. As $|x|$ grows, $\dfrac{1}{x}$ gets closer to 0: asymptote $y = 0$.

Exercise 4.2 · essential.

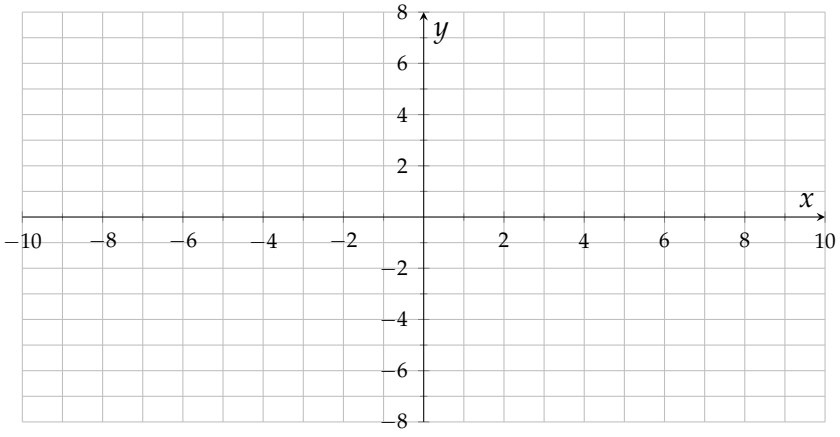
1. Give the domain and the asymptotes of $f(x) = 6 + \dfrac{8}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study the function $g(x) = x + \dfrac{36}{x}$ on $]0, +\infty[$.

Solution. The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 6$. $f'(x) = -\dfrac{8}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{36}{x^2}$ and the minimum is reached at $x = 6$, with value 12.

Further practice: Calao, course pp. 88–93; exercises p. 94, no. 4–9. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. Let h be the function defined on $\mathbb{R}^*$ by $h(x) = -1 + \dfrac{5}{x}$.

1. Compute $h(x)$ for $x \in \{-10{,}000, -1{,}000, -100, 100, 1{,}000, 10{,}000\}$. What can be conjectured about $h(x)$ when x becomes very large in absolute value? Which asymptote can be deduced?
2. Compute $h(x)$ for $x \in \{-0.1, -0.01, -0.001, 0.001, 0.01, 0.1\}$. Describe the behaviour of $h(x)$ when x gets close to 0, first through negative values, then through positive values. Which asymptote can be deduced?
3. Compute $h(-5)$, $h(-1)$, $h(1)$ and $h(5)$, then sketch the curve of h and its asymptotes on the grid below.



4. Application. A training organisation rents a room for 450 € per day and spends 12 €

on meals per participant. For x participants ($x \geqslant 1$), the average cost per participant is $c(x) = 12 + \dfrac{450}{x}$ (in euros).

a. Compute $c(10)$, $c(50)$ and $c(150)$.

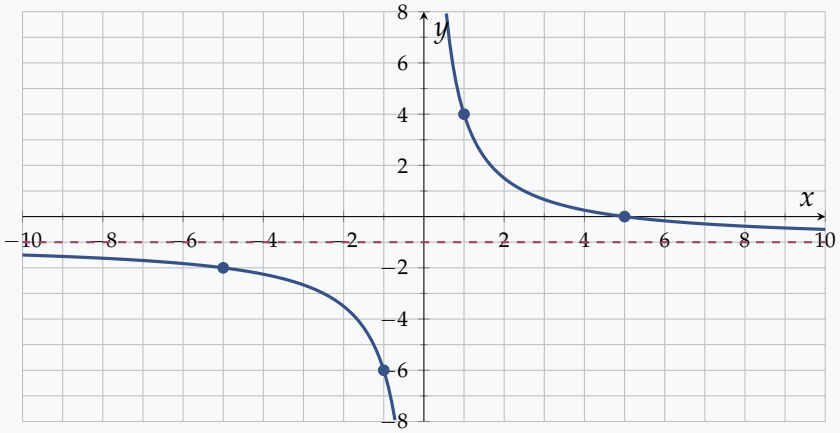
b. Which value does the average cost get closer to when the number of participants becomes very large? Can it fall below 12 €? Justify.

Solution. **Target:** End behaviour and asymptotes.

1. $h(-10{,}000) = -1.000\,5$; $h(-1{,}000) = -1.005$; $h(-100) = -1.05$; $h(100) = -0.95$; $h(1{,}000) = -0.995$; $h(10{,}000) = -0.999\,5$. When $|x|$ becomes very large, $\dfrac{5}{x}$ gets close to 0, so $h(x)$ gets close to -1 (from below at $-\infty$, from above at $+\infty$). The line with equation $y = -1$ is a horizontal asymptote.

2. $h(-0.1) = -51$; $h(-0.01) = -501$; $h(-0.001) = -5{,}001$; $h(0.001) = 4{,}999$; $h(0.01) = 499$; $h(0.1) = 49$. When x gets close to 0 through negative values, $h(x)$ is negative and as large as we like in absolute value ($h(x) \to -\infty$); through positive values, $h(x) \to +\infty$. The line with equation $x = 0$ is a vertical asymptote.

3. $h(-5) = -2$, $h(-1) = -6$, $h(1) = 4$, $h(5) = 0$.



4. a. $c(10) = 12 + 45 = 57$ € ; $c(50) = 12 + 9 = 21$ € ; $c(150) = 12 + 3 = 15$ €.

   b. When x becomes very large, $\dfrac{450}{x}$ gets close to 0: the average cost gets close to 12 € (asymptote $y = 12$). Since $\dfrac{450}{x} > 0$ for $x > 0$, we always have $c(x) > 12$: the average cost never falls below 12 €, as the fixed cost of the room is shared among the participants without disappearing.

Homework. Redo practice exercise 3 with the following data: replace the initial value by 240 and the rate or common ratio by 2. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The answer depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 25 — Derivative and direction of variation

Reciprocal (inverse) function

5 Jan 2027 • B: 2 h – C: 1 h

Objectives.

- Differentiate the reciprocal function.
- Justify the variations.
- Draw up a table.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for an increase of 10%. $1 + \dfrac{10}{100} = 1.10$
2. Differentiate $3x^2 - 5x + 1$. $6x - 5$
3. Solve $(x-2)(x+1) = 0$. $x = -1$ **or** $x = 2$
4. Give the excluded value of $\dfrac{x+1}{x}$. $x = 0$
5. Calculate $\dfrac{1}{7}$ and $\dfrac{1}{-7}$. $\dfrac{1}{7}$ **and** $-\dfrac{1}{7}$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: differentiate the reciprocal function and justify the variations.

Activity 3. Starting from the difference quotient of $f(x) = \dfrac{1}{x}$, recover $f'(x) = -\dfrac{1}{x^2}$. Study its sign and check consistency with the graph.

Solution. The square x^2 is positive for $x \neq 0$, so the derivative is strictly negative on each interval of the domain. Have students say aloud which piece of data triggers the choice of method.

Written record. For $x \neq 0$, $f'(x) = -\dfrac{1}{x^2} < 0$. The reciprocal function decreases on each of the intervals $]-\infty, 0[$ and $]0, +\infty[$.

Example 4. Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Solution. The square x^2 is positive for $x \neq 0$, so the derivative is strictly negative on each interval of the domain.

Exercise 4.3 · essential.

1. Give the domain and the asymptotes of $f(x) = 2 + \dfrac{4}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study the function $g(x) = x + \dfrac{4}{x}$ on $]0, +\infty[$.

Solution. The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 2$. $f'(x) = -\dfrac{4}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{4}{x^2}$ and the minimum is reached at $x = 2$, with value 4.

Further practice: Calao, course pp. 89–93; exercises p. 94, no. 1–3, then p. 96, no. 10–13. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. Compute the derivative of each of the following functions, defined on $\mathbb{R}^*$.

a. $f(x) = 5x + 2 - \dfrac{3}{x}$

b. $g(x) = -4x + 7 + \dfrac{6}{x}$

c. $h(x) = 2x^3 - 5x^2 + \dfrac{1}{x}$

d. $k(x) = \dfrac{1}{2}x^2 + 4x - \dfrac{8}{x}$

e. $p(x) = \dfrac{x^2 + 2}{x}$ (first write $p(x)$ in the form $x + \dfrac{a}{x}$, then give $p'(x)$ as a single quotient)

Solution. **Target:** Differentiate a polynomial plus k/x.

We use: $(ax^n)' = nax^{n-1}$, the derivative of a constant is zero and, for $x \neq 0$, $\left(\dfrac{k}{x}\right)' = k \times \left(-\dfrac{1}{x^2}\right) = -\dfrac{k}{x^2}$.

a. $f'(x) = 5 - \left(-\dfrac{3}{x^2}\right) = 5 + \dfrac{3}{x^2}$.

b. $g'(x) = -4 - \dfrac{6}{x^2}$.

c. $h'(x) = 6x^2 - 10x - \dfrac{1}{x^2}$.

d. $k'(x) = x + 4 + \dfrac{8}{x^2}$.

e. $p(x) = \dfrac{x^2}{x} + \dfrac{2}{x} = x + \dfrac{2}{x}$, so $p'(x) = 1 - \dfrac{2}{x^2} = \dfrac{x^2 - 2}{x^2}$.

Common error: writing $\left(\frac{3}{x}\right)' = \frac{3}{x^2}$, forgetting the "$-$" sign.

Homework. Redo practice exercise 2 with the following data: replace the initial value by 245 and the rate or common ratio by 3. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The answer depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 26 — Combinations with polynomial functions

Reciprocal (inverse) function

7 Jan 2027 • B: 1 h – C: 2 h

Objectives.

- Differentiate a combination of functions.
- Simplify an expression.
- Organise a function study.

Fluency · 6 to 8 minutes, no calculator.

1. Simplify $3x + 8 - x + 2$. $2x + 10$
2. Solve $(x - 2)(x + 1) = 0$. $x = -1$ **or** $x = 2$
3. Give the excluded value of $\dfrac{x+1}{x}$. $x = 0$
4. Calculate $\dfrac{1}{8}$ and $\dfrac{1}{-8}$. $\dfrac{1}{8}$ **and** $-\dfrac{1}{8}$
5. Differentiate $3x^2 - 5x + 1$. $6x - 5$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: differentiate a combination of functions and simplify an expression.

Activity 4. A company produces $x > 0$ hundred items and its average cost is $C(x) = 2x + 30 + \dfrac{200}{x}$. Calculate $C'(x)$, factorise its numerator and find the production level that minimises the average cost.

Solution. $C'(x) = 2 - \dfrac{200}{x^2} = \dfrac{2(x-10)(x+10)}{x^2}$. The minimum is reached for $x = 10$, that is 1,000 items. Have students say aloud which piece of data triggers the choice of method.

Written record. To study $P(x) + \dfrac{k}{x}$, find the domain, calculate the derivative, put over a common denominator, study its sign, then draw up the variations.

Example 5. Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Solution. $C'(x) = 2 - \dfrac{200}{x^2} = \dfrac{2(x-10)(x+10)}{x^2}$. The minimum is reached for $x = 10$, that is 1,000 items.

Exercise 4.4 · essential.

1. Give the domain and the asymptotes of $f(x) = 3 + \dfrac{5}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study the function $g(x) = x + \dfrac{9}{x}$ on $]0, +\infty[$.

Solution. The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 3$. $f'(x) = -\dfrac{5}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{9}{x^2}$ and the minimum is reached at $x = 3$, with value 6.

Further practice: Calao, p. 96, no. 14–21. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. Let f be the function defined on $\mathbb{R}^*$ by $f(x) = 9x - 2 + \dfrac{4}{x}$.

1. Show that, for every non-zero real x, $f'(x) = \dfrac{(3x-2)(3x+2)}{x^2}$.
2. Study the sign of $f'(x)$ on $\mathbb{R}^*$ using a sign table.
3. Deduce the table of variations of f, including the values of $f\left(-\dfrac{2}{3}\right)$ and $f\left(\dfrac{2}{3}\right)$.
4. Compare $f\left(-\dfrac{2}{3}\right)$ and $f\left(\dfrac{2}{3}\right)$. Can a local maximum be less than a local minimum? Explain using the table.

Solution. **Target:** Factorised derivative, sign and variations.

1. $f'(x) = 9 - \dfrac{4}{x^2} = \dfrac{9x^2 - 4}{x^2} = \dfrac{(3x-2)(3x+2)}{x^2}$ (identity $a^2 - b^2 = (a-b)(a+b)$ with $a = 3x$, $b = 2$).
2. $x^2 > 0$ for $x \neq 0$: the sign of $f'(x)$ is that of $(3x-2)(3x+2)$. $3x + 2 = 0 \iff x = -\dfrac{2}{3}$; $3x - 2 = 0 \iff x = \dfrac{2}{3}$.

x	$-\infty$		$-\frac{2}{3}$		0		$\frac{2}{3}$		$+\infty$
$3x+2$		$-$	0	$+$	$\|\|$	$+$		$+$	
$3x-2$		$-$		$-$	$\|\|$	$-$	0	$+$	
x^2		$+$		$+$	0	$+$		$+$	
$f'(x)$		$+$	0	$-$	$\|\|$	$-$	0	$+$	

3. $f\left(-\dfrac{2}{3}\right) = -6 - 2 - 6 = -14$ and $f\left(\dfrac{2}{3}\right) = 6 - 2 + 6 = 10$.

x	$-\infty$		$-\frac{2}{3}$		0		$\frac{2}{3}$		$+\infty$
$f(x)$		$\nearrow$	-14	$\searrow$	$\|\|$	$\searrow$	10	$\nearrow$	

f is increasing on $\left]-\infty;-\frac{2}{3}\right]$, decreasing on $\left[-\frac{2}{3};0\right[$ and on $\left]0;\frac{2}{3}\right]$, increasing on $\left[\frac{2}{3};+\infty\right[$.

4. $-14 < 10$: the local maximum (reached at $-\frac{2}{3}$, on $(-\infty,0)$) is less than the local minimum (reached at $\frac{2}{3}$, on $(0,+\infty)$). This is not a contradiction: they are local extrema on two intervals separated by the forbidden value 0, where the curve "jumps" (vertical asymptote).

Homework. Redo practice exercise 3 with the following data: replace the initial value by 250 and the rate or common ratio by 4. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The answer depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 27 — Sign and factorisation

Reciprocal (inverse) function

12 Jan 2027 • B: 2 h – C: 1 h

Objectives.

- Factorise.
- Build a sign table.
- Respect excluded values.

Fluency · 6 to 8 minutes, no calculator.

1. Calculate 20% of 180. 36
2. Give the excluded value of $\dfrac{x+1}{x}$. $x = 0$
3. Calculate $\dfrac{1}{9}$ and $\dfrac{1}{-9}$. $\dfrac{1}{9}$ **and** $-\dfrac{1}{9}$
4. Differentiate $3x^2 - 5x + 1$. $6x - 5$
5. Solve $(x-2)(x+1) = 0$. $x = -1$ **or** $x = 2$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: factorise and build a sign table.

Activity 5. Study the sign of $\dfrac{(x-2)(x+3)}{x}$. Build the table with the three critical values, distinguish a zero from an excluded value, then solve the inequality $\dfrac{(x-2)(x+3)}{x} \geq 0$.

Solution. The critical values are $-3, 0, 2$. The solution is $[-3, 0[\cup [2, +\infty[$. Have students say aloud which piece of data triggers the choice of method.

Written record. In a quotient, the sign depends on the numerator and the denominator. Excluded values are left out of the table, even if a factor formally cancels.

Example 6. Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Solution. The critical values are $-3, 0, 2$. The solution is $[-3, 0[\cup [2, +\infty[$.

Exercise 4.5 · essential.

1. Give the domain and the asymptotes of $f(x) = 4 + \dfrac{6}{x}$.
2. Calculate $f'(x)$ and study its sign.

3. Study the function $g(x) = x + \dfrac{16}{x}$ on $]0, +\infty[$.

Solution. The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 4$. $f'(x) = -\dfrac{6}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{16}{x^2}$ and the minimum is reached at $x = 4$, with value 8.

Further practice: Calao, pp. 96–101. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. Let h be the function defined on $\mathbb{R}^*$ by $h(x) = 2x - 3 - \dfrac{5}{x}$.

1. Show that, for every non-zero real x, $h'(x) = \dfrac{2x^2 + 5}{x^2}$.
2. Study the sign of $h'(x)$ and deduce the variations of h.
3. Compute $h(-1)$ and $h(1)$. Can we say that "h is increasing on $\mathbb{R}^*$"? Explain.
4. Show that, for every non-zero real x, $h(x) = \dfrac{(2x-5)(x+1)}{x}$. Draw the sign table of $h(x)$, then solve the inequality $h(x) \geqslant 0$.

Solution. **Target:** Always-positive derivative and sign of a quotient.

1. $h'(x) = 2 - \left(-\dfrac{5}{x^2}\right) = 2 + \dfrac{5}{x^2} = \dfrac{2x^2 + 5}{x^2}$.
2. For $x \neq 0$: $2x^2 + 5 \geqslant 5 > 0$ and $x^2 > 0$, so $h'(x) > 0$. The function h is strictly increasing on $(-\infty, 0)$ and strictly increasing on $(0, +\infty)$.

x	$-\infty$		0		$+\infty$
$h'(x)$		$+$	$\|\|$	$+$	
$h(x)$		$\nearrow$	$\|\|$	$\nearrow$	

3. $h(-1) = -2 - 3 + 5 = 0$ and $h(1) = 2 - 3 - 5 = -6$. We have $-1 < 1$ but $h(-1) > h(1)$: h is not increasing on the whole of $\mathbb{R}^*$. It is increasing on each of the two intervals, separately.
4. $\dfrac{(2x-5)(x+1)}{x} = \dfrac{2x^2 - 3x - 5}{x} = 2x - 3 - \dfrac{5}{x} = h(x)$. Critical values: -1, 0 (forbidden value) and $\dfrac{5}{2}$.

x	$-\infty$		-1		0		$\frac{5}{2}$		$+\infty$
$2x-5$		$-$		$-$		$-$	0	$+$	
$x+1$		$-$	0	$+$		$+$		$+$	
x		$-$		$-$	0	$+$		$+$	
$h(x)$		$-$	0	$+$	$\|\|$	$-$	0	$+$	

The solution set of $h(x) \geqslant 0$ is $[-1, 0) \cup \left[\dfrac{5}{2}; +\infty\right[$ (0 is excluded as a forbidden value).

Check: $h(-0.5) = 6 > 0$, $h(1) = -6 < 0$, $h(3) = \dfrac{4}{3} > 0$.

Homework. Redo practice exercise 2 with the following data: replace the initial value by 255 and the rate or common ratio by 5. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The answer depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 28 — Derivatives of polynomials of degree at most 3

Reciprocal (inverse) function

14 Jan 2027 • B: 1 h – C: 2 h

Objectives.

- Differentiate a polynomial.
- Study a derivative.
- Deduce the variations.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for an increase of 25%. $1 + \dfrac{25}{100} = 1.25$
2. Calculate $\dfrac{1}{3}$ and $\dfrac{1}{-3}$. $\dfrac{1}{3}$ **and** $-\dfrac{1}{3}$
3. Differentiate $3x^2 - 5x + 1$. $6x - 5$
4. Solve $(x-2)(x+1) = 0$. $x = -1$ **or** $x = 2$
5. Give the excluded value of $\dfrac{x+1}{x}$. $x = 0$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: differentiate a polynomial and study a derivative.

Activity 6. For $f(x) = 2x^3 - 3x^2 - 12x + 5$, calculate $f'(x)$, factorise, study its sign and deduce the variations of f.

Solution. $f'(x) = 6x^2 - 6x - 12 = 6(x-2)(x+1)$. The function increases on $]-\infty, -1]$, decreases on $[-1, 2]$, then increases. Have students say aloud which piece of data triggers the choice of method.

Written record. $(ax^3 + bx^2 + cx + d)' = 3ax^2 + 2bx + c$. The sign of the derivative gives the direction of variation of the function.

Example 7. Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Solution. $f'(x) = 6x^2 - 6x - 12 = 6(x-2)(x+1)$. The function increases on $]-\infty, -1]$, decreases on $[-1, 2]$, then increases.

Exercise 4.6 · essential.

1. Give the domain and the asymptotes of $f(x) = 5 + \dfrac{7}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study the function $g(x) = x + \dfrac{25}{x}$ on $]0, +\infty[$.

Solution. The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 5$. $f'(x) = -\dfrac{7}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{25}{x^2}$ and the minimum is reached at $x = 5$, with value 10.

Further practice: Calao, pp. 94–96. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. Let f be the reciprocal function, defined on $\mathbb{R}^*$ by $f(x) = \dfrac{1}{x}$.

1. Let h be a non-zero real number, with $h \neq -2$. Show that the rate of change of f between 2 and $2+h$ is $\dfrac{f(2+h) - f(2)}{h} = -\dfrac{1}{2(2+h)}$. Letting h tend to 0, deduce $f'(2)$.
2. We admit that, for every non-zero real x, $f'(x) = -\dfrac{1}{x^2}$. Check consistency with question 1, then compute $f'(-1)$ and $f'(0.5)$.
3. Study the sign of $f'(x)$ and draw the table of variations of f.
4. A student notices that $-2 < 3$ and $f(-2) < f(3)$, and claims that this contradicts the fact that f is decreasing. What should we answer?

Solution. **Target:** Derivative and variations of the reciprocal function.

1. $f(2+h) - f(2) = \dfrac{1}{2+h} - \dfrac{1}{2} = \dfrac{2-(2+h)}{2(2+h)} = \dfrac{-h}{2(2+h)}$. Dividing by $h \neq 0$: $\dfrac{f(2+h) - f(2)}{h} = -\dfrac{1}{2(2+h)}$. When h tends to 0, this rate tends to $-\dfrac{1}{2 \times 2} = -\dfrac{1}{4}$: $f'(2) = -\dfrac{1}{4}$.
2. $-\dfrac{1}{2^2} = -\dfrac{1}{4}$: consistent. $f'(-1) = -\dfrac{1}{(-1)^2} = -1$ and $f'(0.5) = -\dfrac{1}{0.25} = -4$ (slopes of the tangents at these points).
3. For $x \neq 0$, $x^2 > 0$, so $f'(x) = -\dfrac{1}{x^2} < 0$.

x	$-\infty$		0		$+\infty$
$f'(x)$		$-$	‖	$-$	
$f(x)$		↘	‖	↘	

f is strictly decreasing on $(-\infty, 0)$ and strictly decreasing on $(0, +\infty)$.

4. $f(-2) = -0.5$ and $f(3) \approx 0.33$: indeed $f(-2) < f(3)$. There is no contradiction: -2 and 3 are not in the same interval. The reciprocal function is decreasing on each interval separately, not on the whole of $\mathbb{R}^*$; images can be compared using the direction of variation only for two real numbers in the same interval.

Homework. Redo practice exercise 3 with the following data: replace the initial value by 260 and the rate or common ratio by 6. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The answer depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 29 — Full study and graph

Reciprocal (inverse) function

19 Jan 2027 • B: 2 h – C: 1 h

Objectives.

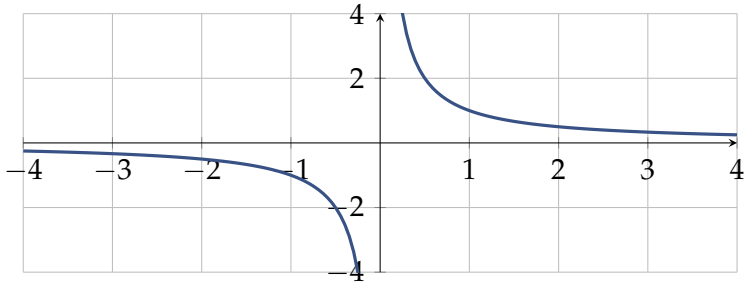
- Carry out a full study.
- Draw a consistent curve.
- Interpret an extremum.

Fluency · 6 to 8 minutes, no calculator.

1. Simplify $3x + 4 - x + 2$. $2x + 6$
2. Differentiate $3x^2 - 5x + 1$. $6x - 5$
3. Solve $(x - 2)(x + 1) = 0$. $x = -1$ **or** $x = 2$
4. Give the excluded value of $\dfrac{x+1}{x}$. $x = 0$
5. Calculate $\dfrac{1}{4}$ and $\dfrac{1}{-4}$. $\dfrac{1}{4}$ **and** $-\dfrac{1}{4}$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: carry out a full study and draw a consistent curve.

Activity 7. Study $g(x) = x + \dfrac{9}{x}$ completely on $]0, +\infty[$: domain, derivative, sign, variations, minimum, behaviour at the boundaries and sketch.



Solution. $g'(x) = \dfrac{(x-3)(x+3)}{x^2}$; minimum $g(3) = 6$; $g(x)$ tends to $+\infty$ at 0^+ and at $+\infty$. Have students say aloud which piece of data triggers the choice of method.

Written record. A full study includes: domain, derivative, sign, variations, limits or behaviour at the boundaries, key points and a consistent graph.

Example 8. Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Solution. $g'(x) = \dfrac{(x-3)(x+3)}{x^2}$; minimum $g(3) = 6$; $g(x)$ tends to $+\infty$ at 0^+ and at $+\infty$.

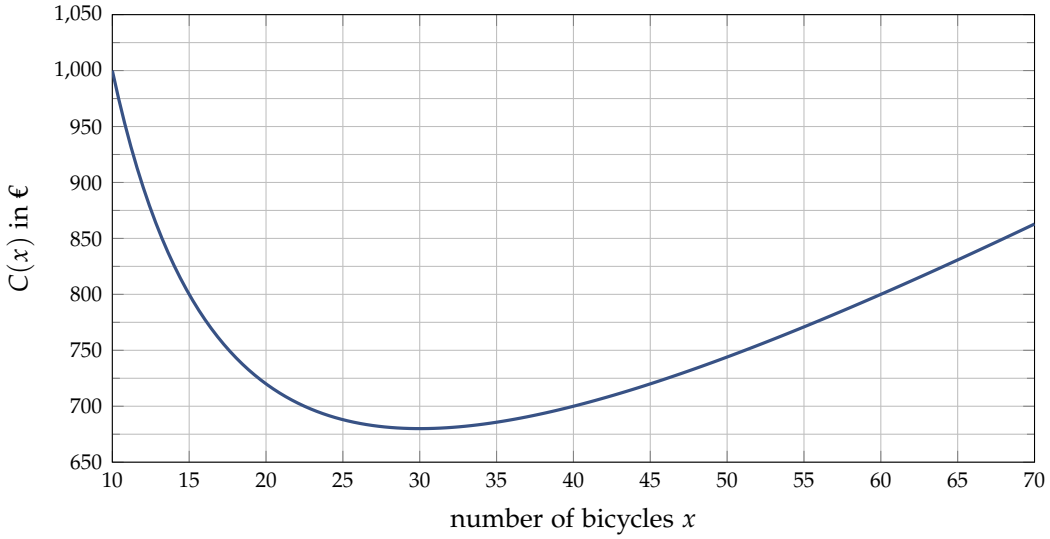
Exercise 4.7 · essential.

1. Give the domain and the asymptotes of $f(x) = 6 + \dfrac{8}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study the function $g(x) = x + \dfrac{36}{x}$ on $]0, +\infty[$.

Solution. The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 6$. $f'(x) = -\dfrac{8}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{36}{x^2}$ and the minimum is reached at $x = 6$, with value 12.

Further practice: Calao, pp. 98–103. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. A company assembles between 10 and 70 electric bicycles per month. For x bicycles assembled, let $C(x)$ be the average unit production cost, in euros. The curve of the function C on $[10, 70]$ is given below.



1. Read the average unit cost when the company assembles 15 bicycles, then 50 bicycles.
2. For which production levels is the average unit cost equal to 800 €?
3. What is the minimum average unit cost? For which production level is it reached?
4. We admit that $C(x) = 8x + 200 + \dfrac{7{,}200}{x}$. Compute $C'(x)$, check that $C'(x) =$

$\dfrac{8(x-30)(x+30)}{x^2}$ and confirm the answer to question 3 by calculation.

Solution. **Target:** Reading an average cost from a graph.

Require reading lines (dotted) and units; a graphical reading is approximate.

1. We read $C(15) \approx 800$ € and $C(50) \approx 745$ € (exact value with the formula of question 4: 744 €).
2. Draw the line with equation $y = 800$: it meets the curve at the points with abscissas 15 and 60. The average cost is 800 € for 15 or 60 bicycles.
3. The lowest point of the curve has coordinates about $(30, 680)$: minimum cost of about 680 € per bicycle, for 30 bicycles per month.
4. $C'(x) = 8 - \dfrac{7{,}200}{x^2} = \dfrac{8x^2 - 7{,}200}{x^2} = \dfrac{8(x^2-900)}{x^2} = \dfrac{8(x-30)(x+30)}{x^2}$. On $[10, 70]$, $x + 30 > 0$ and $x^2 > 0$: $C'(x)$ has the sign of $x - 30$, negative on $[10, 30)$ and positive on $(30, 70]$. C is decreasing on $[10, 30]$ then increasing on $[30, 70]$; its minimum is $C(30) = 240 + 200 + 240 = 680$ €. Checks: $C(15) = 120 + 200 + 480 = 800$ and $C(60) = 480 + 200 + 120 = 800$.

Homework. Redo practice exercise 2 with the following data: replace the initial value by 265 and the rate or common ratio by 7. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The answer depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 30 — **Review and assessment**

Reciprocal (inverse) function — 21 Jan 2027 • B: 1 h – C: 2 h

Objectives.

- Use domain, derivative and sign.
- Write up a study.
- Correct work methodically.

Fluency · 6 to 8 minutes, no calculator.

1. Calculate 5% of 100. 5
2. Solve $(x-2)(x+1)=0$. $x=-1$ **or** $x=2$
3. Give the excluded value of $\dfrac{x+1}{x}$. $x=0$
4. Calculate $\dfrac{1}{5}$ and $\dfrac{1}{-5}$. $\dfrac{1}{5}$ **and** $-\dfrac{1}{5}$
5. Differentiate $3x^2-5x+1$. $6x-5$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: use domain, derivative and sign and write up a study.

Activity 8. Correct a deliberately flawed study of $h(x)=x+\dfrac{4}{x}$: stated domain $\mathbb{R}$, derivative $1+\dfrac{4}{x^2}$, table with no break at 0. Rewrite a complete solution.

Solution. The domain is $\mathbb{R}\setminus\{0\}$ and $h'(x)=1-\dfrac{4}{x^2}$. The table is split at 0 and takes into account -2 and 2. Have students say aloud which piece of data triggers the choice of method.

Written record. Check the four items separately: domain, expression of the derivative, sign of the derivative, variations. An error at one step must not hide the following ones.

Example 9. Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Solution. The domain is $\mathbb{R}\setminus\{0\}$ and $h'(x)=1-\dfrac{4}{x^2}$. The table is split at 0 and takes into account -2 and 2.

Exercise 4.8 · essential.

1. Give the domain and the asymptotes of $f(x) = 2 + \dfrac{4}{x}$.
2. Calculate $f'(x)$ and study its sign.
3. Study the function $g(x) = x + \dfrac{4}{x}$ on $]0, +\infty[$.

Solution. The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 2$. $f'(x) = -\dfrac{4}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{4}{x^2}$ and the minimum is reached at $x = 2$, with value 4.

Further practice: Calao, pp. 100–103. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. A composting site produces x tonnes of compost each week, with $x \in [2, 25]$. The total weekly production cost, in euros, is modelled by $C(x) = 2x^2 + 40x + 288$. Let f be the function giving the average production cost of one tonne: $f(x) = \dfrac{C(x)}{x}$.

1. Show that, for every $x \in [2, 25]$, $f(x) = 2x + 40 + \dfrac{288}{x}$.
2. Compute $f'(x)$ and show that, for every $x \in [2, 25]$, $f'(x) = \dfrac{2(x-12)(x+12)}{x^2}$.
3. Study the sign of $f'(x)$ on $[2, 25]$ and draw the table of variations of f (including the values at the endpoints).
4. What is the minimum average cost of one tonne of compost? For which weekly production is it reached? What is the total cost of the week then?

Solution. **Target:** Minimising an average cost with the derivative.

1. For $x \in [2, 25]$, $x \neq 0$: $f(x) = \dfrac{2x^2}{x} + \dfrac{40x}{x} + \dfrac{288}{x} = 2x + 40 + \dfrac{288}{x}$.
2. $f'(x) = 2 - \dfrac{288}{x^2} = \dfrac{2x^2 - 288}{x^2} = \dfrac{2(x^2 - 144)}{x^2} = \dfrac{2(x-12)(x+12)}{x^2}$.
3. On $[2, 25]$: $2 > 0$, $x + 12 > 0$ and $x^2 > 0$, so $f'(x)$ has the sign of $x - 12$: $f'(x) < 0$ on $[2, 12)$, $f'(12) = 0$, $f'(x) > 0$ on $(12, 25]$. $f(2) = 4 + 40 + 144 = 188$; $f(12) = 24 + 40 + 24 = 88$; $f(25) = 50 + 40 + 11.52 = 101.52$.

x	2		12		25
$f'(x)$		$-$	0	$+$	
$f(x)$	188	$\searrow$	88	$\nearrow$	101.52

4. The minimum average cost is 88 € per tonne, reached for a production of 12 tonnes per week. The total cost is then $C(12) = 288 + 480 + 288 = 1{,}056$ € (check: $12 \times 88 = 1{,}056$).

Homework. Redo practice exercise 3 with the following data: replace the initial value by 270 and the rate or common ratio by 2. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The answer depends on the exercise revisited. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.